Total No. of Questions : 8]

SEAT No. : ____

P502

[Total No. of Pages : 2

[4234] - 102

M.Sc. - I

BOTANY

BO - 1.2 : Plant Physiology and Biochemistry

(2008 Pattern) (Sem. - I)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, taking at least two questions from each section.***
2) ***Answer to the two sections should be written in SEPARATE answer book.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) Give an account on electron transport chain in chloroplast. Add a note on Rubisco Activity.

Q2) Explain the process of glyconeogenesis giving its importance in plants.

Q3) Explain :

a) Mechanism of action of Abscicsic acid.

b) Aquaporins and their role in water transport.

Q4) Write notes on any two :

a) Metabolic changes during seed germination.

b) Biotic stress

c) Uniport and symport channels.

P.T.O.

SECTION - II

Q5) Explain biosynthesis of phenolics.

Q6) What are proteins? Explain different protein structures.

Q7) Explain :

a) Dissociation and association constants.

b) Classification of lipids.

Q8) Write short notes on any two :

a) Synthesis of starch

b) NOD Factor

c) Enzyme kinetics

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Total No. of Questions : 8]

SEAT No. :

P504

[Total No. of Pages : 2

[4234] - 201

M.Sc. - I

BOTANY (Part - I)

BO - 2.1 : Systematics of Vascular Plants

(2008 Pattern) (Sem. - II)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Give comparative account of spore producing organs in pteridophytes. **[16]**

Q2) Compare any two systems of gymnosperm classification with suitable examples. **[16]**

Q3) Describe the following : **[16]**

a) Stelar evolution in pteridophytes.

b) Distribution of gymnosperms in India.

Q4) Write notes on any two of the following : **[16]**

a) Economic importance of gymnosperms.

b) Evolutionary significance of heterosporous Pteridophytes.

c) Pentoxylales.

P.T.O.

SECTION - II

Q5) Enlist orders of subclass Liliopsida. Give salient features of order Liliopsida. **[16]**

Q6) Explain with suitable examples the role of palynology and phytochemistry in angiosperm systematics. **[16]**

Q7) Discuss the following : **[16]**

a) Angiosperms as highly evolved and dominant group of plants.

b) Describe ecades and ecotypes.

Q8) Write notes on <u>any two</u> of the following : **[16]**

a) Concept of family and genus.

b) Describe information of tools and their application in taxonomy.

c) Conservation and utilization of Angiosperm taxonomy.

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Total No. of Questions : 8]

SEAT No. :

P506

[Total No. of Pages : 2

[4234] - 203

M.Sc.

BOTANY

BO - 2.3 : Molecular Biology and Genetic Engineering

(2008 Pattern) (Semester - II)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Describe the mechanism of transcription in prokaryotic cell. Add a note on importance of RNA polymerase. **[16]**

Q2) Write mechanism of initiation, elongation and termination of peptide chain in prokaryotic cell with schematic diagrams. **[16]**

Q3) a) Describe the rolling circle method and theta model of replication. **[8]**

b) Explain chemical, physical and spectroscopic properties of DNA. **[8]**

Q4) Write notes on any two of the following : **[16]**

a) Mechanism of Arabinose operon

b) Organization of promoter in eukaryotic cell

c) Types of DNA damage.

P.T.O.

SECTION - II

Q5) Describe method of Agrobacterium mediated gene transfer. Write it's importance in plant genetic Engineering. **[16]**

Q6) What is DNA sequencing. Describe any two methods of DNA sequencing in details. **[16]**

Q7) a) Describe the structure of 'BAC'. **[8]**

b) Describe the Procedure for 'RAPD'. **[8]**

Q8) Write notes on any two : **[16]**

a) Preparation of plasmid DNA

b) 'Ti' and 'Ri' Plasmid

c) Important characters of plasmid vecters

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Total No. of Questions : 8]

SEAT No. : ____

P508

[Total No. of Pages : 2

[4234] - 302

M.Sc. - II

BOTANY

BO - 3.2 : Environmental Botany & Plant Diversity

(2008 Pattern) (Sem. - III)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat-labelled diagrams must be drawn wherever necessary.***

SECTION - I

Q1) What is population ecology? Enlist and describe any three characters in detail

Q2) Describe and give an acount of phytogeographic regions of India.

Q3) a) Explain ecological stability in ecosystem.

b) Describe various ecological effects of heavy metals.

Q4) Write notes on any two :

a) Indian Bio diversity Act.

b) Acid rain

c) EIA

P.T.O.

SECTION - II

Q5) What is water pollution? Enlist its sources. Explain impact of Eutrophication on water quality.

Q6) What is species diversity? Comment on diversity indices, species richness & abundance alpha and beta diversity.

Q7) a) Describe the selections of plants and microbes in restoration ecology.

b) Explain aesthetic, food and economical value and use of biodiversity.

Q8) Write notes on any two:

a) Forms and structure of communities.

b) Factors affecting bio diversity.

c) CBD.

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Total No. of Questions : 8]

SEAT No. : ____

P509

[Total No. of Pages : 2

[4234] - 303

M.Sc. - II

BOTANY

BO - 3.31 : Phycology - I

(2008 Pattern) (Sem. - III) (Special paper - I)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any FIVE questions, taking at least TWO questions from each section.***
2) ***Answer to the TWO sections should be written in SEPARATE answer books.***
3) ***All questions carry EQUAL marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) Give an outline of classification as per smith and comment on molecular systematics in algae.

Q2) a) Give an account of algae flagella architecture.

b) Comment on algae cell division.

Q3) a) Write briefly on chlorococcales,

b) Give brief account of thallus organization in blue green algae.

Q4) Write short notes on any two of the following :

a) Nostoc,

b) BGA as SCP,

c) Chloroplast variation in green algae.

P.T.O.

SECTION - II

Q5) Give important characters of red algae and comment on its systematics.

Q6) a) Comment on life cycle pattern in brown algae,

b) Write briefly on methods of reproduction in red algae.

Q7) a) Comment on intertidal algae,

b) Give importance of phytoplanktons.

Q8) Write brief notes on any two of the following :

a) Periodicity and succession of algae,

b) Physico chemical properties of water,

c) Lentic and Lotic water algae.

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Total No. of Questions : 8]

SEAT No. : ____

P512

[Total No. of Pages : 2

[4234] - 306

M.Sc. - II

BOTANY

BO - 3.34 : Plant Physiology - I

(2008 Pattern) (Sem. - III) (Special Paper - I)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, taking at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Explain how water deficit affects the growth and metabolism? **[16]**

Q2) Give an account of mechanism of flooding tolerance. **[16]**

Q3) Describe the strategies applied by plants to reduce effects of salt stress. **[16]**

Q4) Write notes on any two of the followings : **[16]**

a) Saline - alkaline and sodic soils.

b) Transgenics for drought stress tolerance.

c) Drought avoidance.

SECTION - II

Q5) Explain the generation of toxic products and scavenging systems developed by plants under stress conditions. **[16]**

Q6) What is photo inhibition? Explain mechanism of UV tolerance. **[16]**

Q7) Explain the strategies applied by plants to overcome ion toxicity. **[16]**

Q8) Write notes on <u>any two</u> of the followings : **[16]**

a) Effects of free radicals on metabolism.

b) Induction of stress by air pollutants.

c) Growth regulators and stress tolerance.

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Total No. of Questions : 8]

SEAT No. :

P514

[Total No. of Pages : 2

[4234] - 308

M.Sc.

BOTANY

BO - 3.36 : Plant Biotechnology - I

(2008 Pattern) (Sem. - III) (Special Paper - I)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***

2) ***Answers to the two sections should be written in separate answer books.***

3) ***All questions carry equal marks.***

SECTION - I

Q1) Define micropropagation. Discuss the advantages of tissue culture technique over conventional methods of crop improvement. **[16]**

Q2) a) Explain the laboratory design of a plant tissue culture laboratory. **[8]**

b) Discuss the protocol and schedule of observation of different types of cultures based on explants in plant tissue culture. **[8]**

Q3) a) How are explants sterilized and established under aseptic conditions? **[8]**

b) What is organogenesis? Distinguish between direct and indirect organogenesis. **[8]**

Q4) Write short notes on any two of the following : **[16]**

a) Applications of somaclonal variations.

b) Factors influencing morphogenesis.

c) Procedure and precautions of stock solutions.

P.T.O.

SECTION - II

Q5) What are transgenic plants? Discuss with suitable example the development of transgenic plant for insect resistance. **[16]**

Q6) a) What are somatic hybrids? Explain their applications in plant biotechnology. **[8]**

b) Explain steps of cryopreservation in plant biotechnology. **[8]**

Q7) a) What is phyto remediation? Discuss it with suitable examples for organic and metal polluted sites. **[8]**

b) Describe the improvement of quality of proteins and carbohydrates by using transgenic plants. **[8]**

Q8) Write short notes on any two of the following : **[16]**

a) Importance of cryopreservation.

b) Mycorrhiza and BGA as biofertilizers.

c) Operation and management of Green house.

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Total No. of Questions : 8]

SEAT No. :

P516

[Total No. of Pages : 2

[4234] - 310

M.Sc. - II

BOTANY

BO - 3.38 : Seed Technology - I

(2008 Pattern) (Sem. - III) (Special Paper - I)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat-labelled diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Explain in detail, the Various physiological and biochemical changes during seed germination. **[16]**

Q2) Give a brief account on the economic importance and integrated management of seed borne diseases. **[16]**

Q3) What is seed dormancy? Mention the types and explain the methods of breaking dormancy? **[16]**

Q4) Write short notes on any two of the following : **[16]**

a) Seed quality characteristics.

b) Quarantine for seed.

c) Chemical composition of seed.

P.T.O.

SECTION - II

Q5) Describe the life of any one pest of stored grain and add a note on damage caused by pest and it's control measures. **[16]**

Q6) Define seed deterioration. Explain the causes and preventive measures of seed deterioration. **[16]**

Q7) Describe the constructional features for a good seed warehouse and add a note on the general principles of seed storage. **[16]**

Q8) Write short notes on <u>any two</u> of the following : **[16]**

a) Relation of insects and plants.

b) Development of microsporangium.

c) Factors affecting seed longevity.

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Total No. of Questions : 8]

SEAT No. : ____

P518

[Total No. of Pages : 2

[4234] - 402
M.Sc. - II
BOTANY
BO-4.2 : Applied Botany
(2008 Pattern) (Sem. - IV)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Enlist seaweeds and explain principles and method of production of seaweeds. **[16]**

Q2) a) Give applications of algae in sewage treatment. **[8]**

b) Explain mass production technology and nutritive value of Spirulina. **[8]**

Q3) a) Enlist fungal organic acids. Add note on citric acid fermentation. **[8]**

b) What are fungal enzymes? Briefly write steps in production of fungal enzymes. **[8]**

Q4) Write notes on any two : **[16]**

a) Myconematicides.

b) Fungal SCP.

c) Fungi in mineral biotechnology and particulate adsorption.

P.T.O.

SECTION - II

Q5) What are superficial and deep mycoses? Add a note on dermatomycoses and aspergillosis. **[16]**

Q6) a) Explain how fungi act as antitumour and antiviral agents. **[8]**

b) Comment on mean, mode and median with suitable examples. **[8]**

Q7) a) Explain ANNOVA with suitable examples. **[8]**

b) What is Regression and Correlation? **[8]**

Q8) Write explanatory notes on any two : **[16]**

a) Search engines.

b) Scope of bioinformatics.

c) Methods of sequence analysis.

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Total No. of Questions : 8]

SEAT No. :

P520

[Total No. of Pages : 2

[4234] - 404

M.Sc.

BOTANY

BO - 4.42 : Mycology and Plant Pathology - II

(2008 Pattern) (Sem. - IV) (Special - II)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Attempt a total of five questions from the following, select atleast two questions from each section.***
2) ***Answers to the questions from each section should be written in separtate answer books.***
3) ***Figures to the right indicate full marks.***
4) ***Neat labelled diagrams must be drawn wherever necessary.***

SECTION - I

Q1) What are different methods of fermentation technology? Give detailed account of production of antibiotics. **[16]**

Q2) a) How fungi act as immunoregulator? **[8]**

b) Explain role of fungi in control of weeds and nematodes. **[8]**

Q3) a) Enlist six fungal enzymes. Add a note on steps in industrial production of enzymes. **[8]**

b) Discuss the role of fungi in food industry. **[8]**

Q4) Write explanatory notes on any two : **[16]**

a) White rot fungi in bioremediation.

b) Fungi in Homoeopathy and Ayurvedic medicines.

c) Mycofungicides and Mycoinsecticides.

P.T.O.

SECTION - II

Q5) Give difference between subcutaneous and systemic mycoses. Explain mycetoma and its clinical aspects. **[16]**

Q6) a) Comment on physiology of diseased plants. **[8]**

b) Give brief account of effect of environment on disease development. **[8]**

Q7) a) Give disease cycle of downy mildew and add a note on control measures. **[8]**

b) Comment on white rust disease. **[8]**

Q8) Write explanatory notes on <u>any two</u> : **[16]**

a) Structural and Biochemical defence mechanisms.

b) Biotechnology and its role in plant pathology.

c) Contributions of E. J. Butler and Millardet.

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Total No. of Questions : 8]

SEAT No. : ____

P522

[Total No. of Pages : 2

[4234] - 406

M.Sc. - II

BOTANY

BO - 4.44 : Plant Physiology - II

(2008 Pattern) (Semester - IV) (Special Paper - II)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Describe biosynthesis, degradation and role of carotenoids. **[16]**

Q2) a) Explain photochemical reaction. **[8]**

b) Comment on recent research in India and abroad on climate change and productivity. **[8]**

Q3) a) Explain effect of elevated levels of CO_2 and O_2 on net assimilation rate. **[8]**

b) Give the causes of global warming and its effect on crop yield. **[8]**

Q4) Write notes on any two of the following : **[16]**

a) Ozone layer depletion

b) Pigment organisation in thylakoid membrane

c) Chlorophyll degradation

P.T.O.

SECTION - II

Q5) What are allelochemicals? Explain how they help in plant competition. **[16]**

Q6) a) Explain effect of fungal infection on plant metabolism. **[8]**

b) Explain how, viruses interact with the plants. **[8]**

Q7) a) What are different defence mechanism in host plants? Explain. **[8]**

b) Explain what is hypersensitive response. **[8]**

Q8) Write notes on <u>any two</u> of the following : **[16]**

a) Bt. Cotton

b) Photoreceptors

c) Circadian rythums

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Total No. of Questions : 8]

SEAT No. : ____

P523

[Total No. of Pages : 2

[4234] - 407

M.Sc.

BOTANY

BO-4.45 : Genetics, Molecular Biology and Plant Breeding - II

(2008 Pattern) (Special Paper - II) (Semester - IV)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Attempt five questions with at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***Draw neat labelled diagrams wherever necessary.***
4) ***All questions carry equal marks.***

SECTION - I

Q1) Explain technique of chromosome microdissection and microcloning. **[16]**

Q2) a) Explain process of RFLP. **[8]**

b) Discuss gene-environment interaction. **[8]**

Q3) a) How is yield of DNA measured ? **[8]**

b) Comment on map based cloning. **[8]**

Q4) Write in brief on any two of the following : **[16]**

a) Application of DNA amplification.

b) Colony and plaque hybridization.

c) Chromosome walking.

P.T.O.

SECTION - II

Q5) What are abiotic stresses? How are these characterized? **[16]**

Q6) a) Explain drought resistance mechanisms in plants. **[8]**

b) Write on drought tolerant cultivated varieties of crops. **[8]**

Q7) a) How the vitamin content is enhanced in crop ? Add a note on enhancement of vitamin A. **[8]**

b) What is protein quality? Explain. **[8]**

Q8) Write in brief on any two of the following : **[16]**

a) Inter specific hybridization.

b) Somaclonal variations.

c) Domestication.

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Total No. of Questions : 8]

SEAT No. :

P524

[Total No. of Pages : 2

[4234] - 408

M.Sc.

BOTANY

BO - 4.46 : Plant Biotechnology - II (Special Paper - II)

(2008 Pattern) (Sem. - IV)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Explain functioning of PCR giving its principle and applications. **[16]**

Q2) a) Explain genomics with reference to biotechnology. **[8]**

b) Give the principle and method of western blotting. **[8]**

Q3) a) Describe various types DNA polymorphisms. **[8]**

b) Explain mode of action of any four enzymes used in recombinant DNA technology. **[8]**

Q4) Write explanatory notes on any two of the following : **[16]**

a) Shot gun sequencing.

b) Chromosome walking.

c) Functional genomics.

P.T.O.

SECTION - II

Q5) Explain the mechanism of biological nitrogen fixation. **[16]**

Q6) a) Give applications of plant biotechnology in agriculture with suitable examples. **[8]**

b) Explain role of microbes in leaching of metals. **[8]**

Q7) What is Proteomics? Explain in detail any two procedures. **[16]**

Q8) Write explanatory notes on <u>any two</u> of the following : **[16]**

a) Plasmid vectors.

b) Genome annotations.

c) Phytoremediation - Role in environment protection.

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Total No. of Questions : 8]

SEAT No. :

P525

[Total No. of Pages : 2

[4234] - 409

M.Sc.

BOTANY

BO-4.47 : Plant Biodiversity (Special Paper - II)

(2008 Pattern) (Sem. - IV)

Time : 3 Hours] ***[Max. Marks :80***

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Explain the causes of loss of species extinction Add a note on the process responsible for species extinction. **[16]**

Q2) What is ex-situ conservation. Add a note on the types of ex-situ conservation. **[16]**

Q3) Comment on - **[16]**

a) Role of UNESCO and ISBI in plant biodiversity management.

b) Biodiversity Act.

Q4) Write notes on any two : **[16]**

a) Reasons for loss in diversity of major ecosystems of the world.

b) Concept of sustainable development.

c) Sacred grooves.

P.T.O.

SECTION - II

Q5) Explain the methodologies for evaluation of biodiversity. Add a note on biodiversity prospecting. **[16]**

Q6) Define biological invasions and comment on its ecological and economic impacts. Add a note on role of biotechnology in assessment of biodiversity and bioresources. **[16]**

Q7) Explain : **[16]**

a) Role of metadatabases, virtual libraries in management of plant biodiversity.

b) Ramsar convention.

Q8) Write notes on any two of the following : **[16]**

a) Ecotourism and agroforestry.

b) CBD.

c) Impacts of biotechnology on biodiversity.

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Total No. of Questions : 8]

SEAT No. :

P526

[Total No. of Pages : 2

[4234] - 410

M.Sc. - II

BOTANY

BO - 4.48 : Seed Technology - II (Special Paper - II)

(2008 Pattern) (Semester - IV)

Time : 3 Hours] *[Max. Marks :80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Give an brief account of seed production of maize and soyabean. **[16]**

Q2) Describe multiplication and storage of clones in potato. Add a note on true potato seeds (TPS) production. **[16]**

Q3) Explain post-harvest operations and preparation of seeds for processing. **[16]**

Q4) Write short notes on any two of the following : **[16]**

a) Seed Village concept.

b) Production and maintenance of nucleus.

c) Electrical conductivity separations.

P.T.O.

SECTION - II

Q5) Describe methods and working of Equipments used in seed treatment. **[16]**

Q6) Enlist major organisations involved in seed testing. Explain sampling methods and functioning of equipments used for seed testing. **[16]**

Q7) Give concept and procedure of artificial seed production. **[16]**

Q8) Write a short note on any two of the following : **[16]**

a) Central seed committee and their functioning.

b) PCR and southern hybridization.

c) Seed certification.

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Total No. of Questions : 8]

SEAT No. :

P501

[Total No. of Pages : 2

[4234] - 101
M.Sc. - I
BOTANY
BO - 1.1 : Systematics of Non Vascular Plants
(2008 Pattern) (Sem. - I)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any FIVE questions, taking at least Two questions from each section.***
2) ***Answers to the TWO sections should be written in SEPARATE answer books.***
3) ***All questions carry EQUAL marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) Comment on range of thallus organization in chlorophyta and write characters of life cycle pattern in order charales. **[16]**

Q2) Draw and describe external morphology and internal structure of sporophyte of order Anthocerotales. **[16]**

Q3) Write short answers of the following : **[16]**

a) Asexual reproduction in cyanophyta.

b) Position of algae in eight kingdom system.

Q4) Write short notes on any two of the following : **[16]**

a) Evolutionary trends among algae.

b) Characters of Chrysophyta.

c) Sporophyte of Polytrichum.

P.T.O.

SECTION - II

Q5) Give an account of thallus structure, spore producing structure and life cycle pattern in Myxomycetes. **[16]**

Q6) Give an outline classification of fungi as per by Smith and add a note on Biotrophs. **[16]**

Q7) Write short answers of the following : **[16]**

a) Describe parasexuality in fungi.

b) Comment on asexual reproduction in Mucorales.

Q8) Write short notes on any two of the following : **[16]**

a) Basidiocarp.

b) Spherocarpales.

c) Vegetative reproduction in Bryophytes.

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Total No. of Questions : 8]

SEAT No. :

P503

[Total No. of Pages : 2

[4234] - 103

M.Sc. - I

BOTANY

BO - 1.3 : Principles of Genetics and Plant Breeding

(2008 Pattern) (Semester - I)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least Two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Define quantitative and qualitative inheritance. Describe inheritance of Quantitative traits in Nicotiana and Zea mays.

Q2) Explain how ordered and unordered data is used for gene mapping in Neurospora.

Q3) Give an account of :

a) Complementary factors.

b) Multiple factor hypothesis.

Q4) Write notes on **any two** of the following :

a) C value paradox.

b) Chloroplast genome.

c) Multiple factor hypothesis.

P.T.O.

SECTION - II

Q5) What are chromosomal abberations? Describe different types of chromosomal abberations.

Q6) Give classification of mutation. Explain molecular basis of gene mutations.

Q7) Comment on :

a) Pre and Post Mendelian development.

b) Importance of genetic diversity in crop improvement.

Q8) Write short notes on :

a) Heterosis.

b) Types of incompatability.

c) Role of polyploidy in crop improvement.

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Total No. of Questions : 8]

SEAT No. :

P505

[Total No. of Pages : 2

[4234] - 202

M.Sc.

BOTANY

BO - 2.2 : Cell Biology and Instrumentation

(2008 Pattern) (Semester - II)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Give an account of the evolution of eukaryotic cell from prokaryotic cells. **[16]**

Q2) a) Explain biogensis of cell wall. **[8]**

b) Describe ultra structure and functions of endoplasmic reticulum. **[8]**

Q3) a) Organization of chromosomes. **[8]**

b) Lysosomes are known as suicide bags explain. **[8]**

Q4) Write explanatory notes on any two of the following : **[16]**

a) Dosage compensation.

b) Role of cyclins in mitosis.

c) Glyoxy somes.

P.T.O.

SECTION - II

Q5) Explain different types of microscopes studied by you. **[16]**

Q6) a) Distinguish SEM and TEM. **[8]**

b) Applications of ultracentrifuge. **[8]**

Q7) a) Explain principles and working of ESR spectroscopy. **[8]**

b) Functions of threonine kinase. **[8]**

Q8) Write explanatory notes on any two of the following : **[16]**

a) Immunodiffusion.

b) Microtomy

c) Ion exchange chromatography

■■■

Total No. of Questions : 8] **SEAT No. :**

P507 **[Total No. of Pages : 2**

[4234] - 301
M.Sc. - II
BOTANY
BO - 3.1 : Developmental Botany and Plant Tissue Culture
(2008 Pattern) (Semester - III)

Time : 3 Hours] ***[Max. Marks : 80***

Instructions to the candidates:

1) ***Answer any five questions, taking at least Two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Give an account of intrinsic and extrinsic factors controlling the plant development.

Q2) Explain :

a) Cytochemical changes in vegetative plant body leading vegetative to reproductive phase.

b) Stages of anther development.

Q3) Comment on :

a) Female game to Phyte

b) Developmental routes to parthenocarpy.

Q4) Write short notes on any two of the following :

a) Hormone signaling during growth and development.

b) Expression of self incompatability.

c) Applications of developmental botany.

P.T.O.

SECTION - II

Q5) What is somaclonal variation? Discuss types, causes and applications of somaclonal variations.

Q6) Give an account of applications of plant tissue culture in production of secondary metabolites.

Q7) a) What is meant by totipotency ? Explain the factors affecting to tipotency.

b) Explain the protocol of micropropagation.

Q8) Write short notes on <u>any two</u> of the following :

a) Role of plant growth regulators.

b) Transgenic plants.

c) Isolation of protoplasts.

■■■

Total No. of Questions : 8]

SEAT No. :

P510

[Total No. of Pages : 2

[4234] - 304

M.Sc. - II

BOTANY

BO - 3.32 : Mycology and Plant Pathology - I

(2008 Pattern) (Special Paper - I) (Semester - III)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any FIVE questions, taking to least TWO questions from each section.***
2) ***Answers to the TWO sections should be writtern in SEPARATE answer books.***
3) ***All questions carry EQUAL marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) What are Deuteromycotina? Give detail account of conidiomata and conidia morphology in the Deuteromycotina. **[16]**

Q2) Write short answers of the following : **[16]**

a) "Laboulbeniomycetes is interesting group". Amplify.

b) Give Bessey's classification of fungi.

Q3) Give characters of zygomycotina. Discuss sporangia to conidial evolution in Mucorales. **[16]**

Q4) Write short notes on **any two** of the following : **[16]**

a) Plasmodiophoromycetes

b) Forms of Lichen

c) Aphylloporales

d) Fructifications in Gasteromycetes.

P.T.O.

SECTION - II

Q5) Discuss in detail colonization strategies in fungi. Add a note on soil fungi. **[16]**

Q6) Write short answers of the following : **[16]**

a) Comment on rhizosphere fungi.

b) Briefly write on heterothallism.

Q7) Discuss different aspects of pathogenecity host resistance and virulence. How fungi are ideal organisms for genetical studies? **[16]**

Q8) Write short notes on **any two** of the following : **[16]**

a) Fungal toxins.

b) Fungal nitrogen nutrition.

c) Fungal association with plants.

d) Air borne fungi.

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Total No. of Questions : 8]

SEAT No. :

P511

[Total No. of Pages : 2

[4234] - 305
M.Sc. (Part - II)
BOTANY
BO - 3.33 : Angiosperms
(2008 Pattern) (Semester - III) (Special Paper - I)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, taking at least TWO questions from each section.***
2) ***Answers to the TWO sections should be written in SEPARATE answer books.***
3) ***ALL questions carry EQUAL marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) Discuss the role of Modern trends in systematics of Amentiferae and santalaceae.

Q2) Discuss :

a) Objectives and functions of botanical gardens.

b) Exocarpus and Mesembryanthemum.

Q3) Explain an organization, units, objectives and functions of herbarium.

Q4) Write notes on any two :

a) Mughal gardens.

b) ICBN

c) Procedures of Biosystematic investigation.

P.T.O.

SECTION - II

Q5) Explain floristic composition of India. Add a note on biodiversity of world.

Q6) Describe the following :

a) Clausen's experiment.

b) Effective and valid Publication.

Q7) Give aims and objectives of Biosystematics. Add a note on Numerical taxonomy.

Q8) Write notes on <u>any two</u> :

a) Primitive features of centrospermae.

b) Digital herbarium.

c) Distinguishing features of Loranthoideae and viscoideae.

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Total No. of Questions : 8]

SEAT No. :

P513

[Total No. of Pages : 2

[4234] - 307

M.Sc. -II

BOTANY

BO-3.35 : Genetics, Molecular Biology and Plant Breeding-I (2008 Pattern) (Semester - III) (Special Paper - I)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, taking at least Two questions from each section.***
2) ***Answers to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Explain structural and organizational complexity of eukaryotic chromosome.

Q2) a) Comment on any two methods of bacterial recombination.

b) Write briefly on inheritance of quantitative characters with suitable example.

Q3) a) Explain co-relation of genetic and physical maps.

b) Write an account of karyotype evolution.

c) Give an account of alien gene transfer in wheat through chromosome manipulation.

Q4) Write notes on **any two** of the following :

a) Null hypothesis

b) Haploid - origin and production

c) Gene targeting

P.T.O.

SECTION - II

Q5) Explain population improvement through mass, progeny and recurrent selection.

Q6) Describe the method of handling of material treated with the mutagens and the progeny of M1, M2 and M3 generations.

Q7) a) Write an account of completely randomized block designs.

b) Give characteristics of monosomics and trisomics.

Q8) Write notes on any two of the following :

a) Biochemical genetics

b) Bionomial expansion

c) Production of hybrid seeds.

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Total No. of Questions : 8]

SEAT No. :

P515

[Total No. of Pages : 2

[4234] - 309
M.Sc. - II
BOTANY
BO - 3.37 : Plant Diversity
(2008 Pattern) (Special Paper - I) (Semester - III)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any FIVE questions taking at least two questions from each section.***
2) ***Answers to the TWO sections should be written in SEPARATE answer books.***
3) ***All questions carry EQUAL Marks.***
4) ***Neat diagrams must be drawn WHEREVER necessary.***

SECTION - I

Q1) Comment on global distribution of biodiversity and add a note on endemism and diversity.

Q2) a) Define micro and macro evolution. Add a note on Darwin's evidence for natural selection.

b) Give comparison of different sampling techniques of biodiversity.

Q3) a) Describe fresh water wetlands.

b) Explain marine ecosystem in detail.

Q4) Write short notes on any two :

a) Determinants of genetic diversity.

b) Diversity in domesticated species.

c) Distribution of higher plant species diversity.

P.T.O.

SECTION - II

Q5) Comment on algal and bryophyte diversity with reference to habit, habitat and distribution.

Q6) Explain :

a) Levels of biodiversity.

b) Global distribution of species richness.

Q7) Comment on :

a) Species inventory.

b) Origin and evolution of cultivated species diversity.

Q8) Write notes on <u>any two</u> :

a) Microbial diversity.

b) Sampling techniques for biodiversity assessment.

c) Hotspots in India.

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Total No. of Questions : 8]

SEAT No. :

P517

[Total No. of Pages : 2

[4234] - 401

M.Sc.

BOTANY

BO - 4.1 : Plant Resources and Evolution

(2008 Pattern) (Semester - IV)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Attempt a total of five questions from the following, selecting at least two questions from each section.***
2) ***Answers to the questions from each section should be written in seperate answer books.***
3) ***Figures to the right indicate full marks.***
4) ***Neat labelled diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Describe two resins and two fibre yielding crops w.r.t. botanical name, part used, chemical constituents and therapeutic uses. **[16]**

Q2) a) Comment on the methods of phytochemical investigation w.r.t. secondary metabolites. **[8]**

b) Enlist and describe Vavilov's center of origin. **[8]**

Q3) a) Give the significance of energy plantations . **[8]**

b) What are secondary metabolites? Give a brief account of types of secondary metabolites present in medicinal plants with one example each **[8]**

Q4) Write explanatory notes on **any two** of the following : **[16]**

a) Miller's experiment.

b) Major events in evolutionary time scale.

c) Concept of variation.

P.T.O.

SECTION - II

Q5) Give monographic account of any one drug obtained from bark and rhizome. **[16]**

Q6) a) What is chemotaxonomy? Give its role in criminology. **[8]**

b) Comment on microscopic evaluation for standardization of crude drugs. **[8]**

Q7) a) Write in brief on molecular tools in phylogeny. **[8]**

b) Explain sexual selection in the mechanism of evaluation. **[8]**

Q8) Write explanatory notes on any two of the following : **[16]**

a) Importance of ethnobotany.

b) Cordaitales.

c) Spontaneity of mutations.

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Total No. of Questions : 8]

SEAT No. :

P519

[Total No. of Pages : 2

[4234] - 403
M.Sc. - II
BOTANY
BO - 4.41 : Phycology
(Special Paper - II) (2008 Pattern) (Semester - IV)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) Describe mass Production technology of Spirulina. comment on its nutritional value. **[16]**

Q2) a) Describe different methods used in isolation of algae. **[8]**

b) Explain in brief strain selection, and measurement of growth in culturing of algae. **[8]**

Q3) a) Describe processing and quality standards of large scale cultivation of algae. **[8]**

b) Give an account of different media used in culturing of algae. **[8]**

Q4) Write short notes on any two of the following : **[16]**

a) Nutritional value of Chlorella

b) Agarophytes

c) Maintenance of algal cultures.

P.T.O.

SECTION - II

Q5) What is bioremidiation? Explain the role of algae in sewage disposal and waste water treatment. **[16]**

Q6) a) Describe various types of biofertilizers and comment on seaweed liquid fertilizer. **[8]**

b) Comment on the various seaweed resources of India. **[8]**

Q7) a) Explain in brief the role of algae in biotechnology. **[8]**

b) Describe the use of algae in production of biofuel. **[8]**

Q8) Write short notes on <u>any two</u> of the following : **[16]**

a) SCP

b) Tissue culture of marine Macroalgae

c) Algal biotechnology and enterpreneurship development.

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Total No. of Questions : 8]

SEAT No. :

P521

[Total No. of Pages : 2

[4234] - 405
M.Sc. (Part - II)
BOTANY
BO-4.43 : Angiosperms - II
(2008 Pattern) (Special - II) (Semester - IV)

Time : 3 Hours] *[Max. Marks : 80*

Instructions to the candidates:

1) ***Answer any five questions, selecting at least two questions from each section.***
2) ***Answer to the two sections should be written in separate answer books.***
3) ***All questions carry equal marks.***
4) ***Neat diagrams must be drawn wherever necessary.***

SECTION - I

Q1) What are avenues? Explain criteria for selection of trees for road-sides and public gardens. **[16]**

Q2) Explain :

a) Somatic embryogenesis **[8]**

b) Structure and distribution of wood elements. **[8]**

Q3) Enlist major timber yielding plants. Give distribution of wood elements and uses of any two timber plants. **[16]**

Q4) Write explanatory notes on any two of the following : **[16]**

a) After care of trees.

b) VAM application.

c) Arborescent monocotyledons.

P.T.O.

SECTION - II

Q5) Describe ultra structure of pollen. Add a note on its biochemical aspects. **[16]**

Q6) Explain foraging behaviour of bees in relation to pollen. Add a note on floral calender. **[16]**

Q7) a) Describe pollen germination in vivo and in vitro. **[8]**

b) Explain the steps involved in androgenesis. **[8]**

Q8) Write explanatory notes on any two of the following : **[16]**

a) Apomixis

b) Causes and significance of polyembryony.

c) Pollen storage.

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